

Scenario 6

A university's security system detects that a student's laptop at IP address 10.50.20.113 has been infected with malware and is attempting to scan other devices on the network. A rapid response is needed.

Question: Describe the immediate action a security application on a centralized SDN controller could take to quarantine the infected device.

Answer:

A security application running on a centralized SDN controller can execute automated, near-instantaneous quarantine actions across the data plane using several methods:

- **1. Direct Drop Flow Rule Installation:**

The security application programmatically instructs the SDN controller via Northbound APIs to construct a high-priority flow rule matching source IP 10.50.20.113 with an action of DROP. The controller immediately pushes this rule via its Southbound API (e.g., OpenFlow) to the flow table of the ingress switch port where the infected laptop is attached, completely isolating it from sending packets to any other network destination.

- **2. Dynamic Redirection to a Remediation VLAN / Sandbox:**

Instead of completely dropping traffic, the controller can push flow rules that match IP 10.50.20.113 and modify its packet actions to rewrite the VLAN tag or alter destination ports. This redirects all outgoing traffic from the laptop to an isolated quarantine VLAN or honeypot/walled garden, allowing IT staff to push remediation instructions to the user.

- **3. Dynamic Physical Port Disablement / Isolation:**

Using OpenFlow port modification commands, the SDN controller can instruct the edge switch to shut down or isolate the specific physical switch port connected to IP 10.50.20.113, stopping malicious activity at Layer 1/2 without affecting any other devices on the switch.

- **Key Advantage of the SDN Approach:**

Because the SDN controller operates programmatically, these isolation actions occur automatically in milliseconds upon malware detection—eliminating manual administrative intervention, stopping lateral malware propagation across the university network instantly.